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Engineering Design Innovation (30.007)

30.007 Term Paper

***Design of a Low-Profile Mecanum Drive AMR
for Trolley Transport***

Submitted By:

Group 10

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ABSTRACT

This report presents the design, modelling, and prototyping of a four-wheel mecanum drive Autonomous Mobile Robot (AMR) intended for indoor textile logistics and material handling applications. The system utilises four BGM 80mm flange 120W brushless DC (BLDC) motors with 15:1 planetary gearboxes, driven by BLD-120A controllers, mounted on a modular aluminium 2020 extrusion chassis measuring 800mm × 500mm. The mecanum wheel configuration (4× 152mm) enables omnidirectional movement, allowing the robot to navigate tight spaces without turning. Key design decisions including chassis structural analysis, thermal management, battery sizing, and component placement, are discussed. Finite element analysis and classical beam theory were used to predict frame deflection under payload, with results showing the chassis can support up to 15–20 kg distributed load, extendable to 70 kg with a single cross brace. The motor mounting bracket design using a D1/D2 extension system was validated with a safety factor exceeding 16× at all joints. This paper details the system requirements, concept selection process, modelling approaches, and prototype construction.

I. INTRODUCTION

Autonomous Mobile Robots (AMRs) are increasingly deployed in warehouses, factories, hospitals, and retail environments to automate material transport. Unlike Automated Guided Vehicles (AGVs) that follow fixed paths, AMRs can navigate dynamically using onboard sensors and compute, adapting to changing environments in real time. The global AMR market is projected to grow significantly as labour costs rise and the demand for flexible automation increases.

Despite this growth, commercially available AMR platforms remain expensive, often exceeding SGD 10,000–50,000 for a basic unit, putting them out of reach for small businesses, research labs, and educational institutions. There is a clear gap for a low-cost, modular AMR platform that can be built from readily available components and customised for specific use cases. Solutions like the Flexley Mover® P603 platform AMR and the Seagrid tow tractor S7 AMR are currently be used for the above-mentioned purposes, but they face a critical operational constraint: their inability to fit into narrow, tight spaces, specifically directly underneath a transport trolley.

This project addresses these gaps by developing a low-profile, four-wheel mecanum drive AMR. The mecanum wheel configuration provides omnidirectional capability, which is critical for navigating narrow aisles. To interface with the trolley, the proposed design integrates a custom belt-driven extension system with a 90° pivoting clamp end-effector, allowing the robot to seamlessly drive beneath a trolley, deploy outwards, and rigidly lock onto the chassis structure for secure transit.

II. PROPOSED APPROACH

A. System Requirements

The system requirements were derived from the target application of indoor material transport in a small warehouse or lab setting. The key requirements are summarised in Table 1.

Functional Requirement	Target Value	Reasoning
AMR height must fit under trolley	≤ 200 mm	Robot can fit under trolley
Clamping arms must withstand holding force	≥ 300 N	Clamp holds load without malfunction/damage
AMR total mass	≥ 25 kg	Drive wheels maintain grip without slipping under load
Clamp vertical reach	≥ 200 mm	Clamp reaches trolley contact point
AMR must move when wheels/castors are misaligned	Works for all 4 castor orientations	Successful movement from worst-case wheel alignment
Able to move	≤ 100 kg	Trolley + Load of a maximum 100 kg
Travel speed under load	≥ 0.3 m/s, target 0.5–0.7 m/s	Standard AMR speed
Time taken for AMR to clamp and dock on trolley	≤ 1 min	Process is quick and efficient enough

TABLE 1: System Requirements

B. Concept Selection

Our initial objective was dual-faceted, to simultaneously mitigating the dusty environments of textile factories and optimizing trolley mobility. Early design iterations explored active filtration systems, alongside an under-trolley robotic mover base. However, considering system complexity, strict time constraints as well as cost, we pivoted to focus exclusively on the mobility and docking system of the robotic mover. By evaluating options against system requirements using a Pugh chart, we selected a compact platform tailored for narrow aisles and automatic under-trolley coupling.

Criteria	Weight	Current Way Worker Push (REF)	Low Profile Mecanum Drive AMR	Fish-Inspired Conical Rib-Vortex Filter	RFID-BLE Tracking	Air Curtain Trolley
Existing trolley compatibility	5	0	+	0	+	-
Reduce worker effort	5	0	+	+	+	0
Reduce trolley/cover damage	5	0	+	0	0	+
Dust / lint control around trolley	4	0	+	+	0	+
Ease of implementation	5	0	+	+	0	-
Safety during operation	5	0	+	0	0	0
Maintenance ease	4	0	0	+	0	-
Cost and technical feasibility	4	0	0	0	0	-
Weighted Net Score (MAX +37)		0	+29	+18	+10	-9

SELECTED CONCEPT
Low-Profile Mecanum Drive AMR

Figure 1. Ideation Evaluation Pugh Chart

C. Concept Description

The AMR architecture is divided into three primary integrated systems: (1) the structural chassis and drivetrain assembly, (2) the synchronous belt extension and clamping mechanism, and (3) the electronics and motion control system.

1) Structural Chassis & Drivetrain Assembly

The chassis frame is constructed from modular 2020 Series Slot 6 aluminium extrusion, forming a low-profile single-level rectangular footprint measuring 800 mm × 500 mm. The structure is rigidly joined using triangle gussets and L-brackets, featuring a suspended sub-frame shelf dropped 80 mm below the main frame to house the battery, lowering the centre of gravity for dynamic stability.

Locomotion is handled by four corner-mounted 120W brushless DC (BLDC) motors integrated with 15:1 planetary gearboxes, directly driving four 152 mm omnidirectional mecanum wheels. Each motor is secured using a custom D1/D2 double-rail extension bracket lined with triangle gussets to counter high breakaway torque reaction forces when moving heavy loads. This wheel-and-chassis geometry establishes a final structural ground clearance of approximately 22 mm, allowing the low-profile unit to drive seamlessly beneath the cargo trolley.

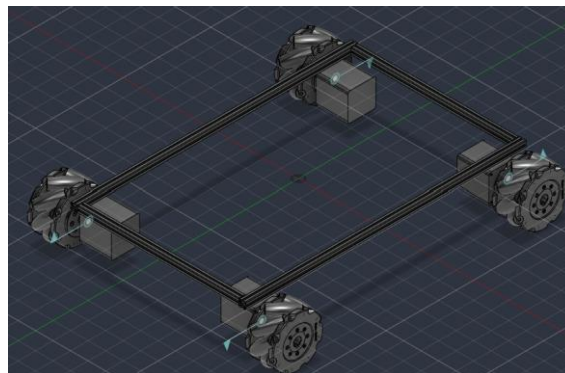


Figure 2: Chassis and Motor Holder CAD Diagrams

2) Synchronous Belt Extension & Clamping System

A belt-driven linear extension system mounted to the upper frame deploys two opposing arms outward to engage the trolley. Each arm is driven by a motor-and-HTD3M-pulley arrangement along guide rails, controlled by a dedicated ESP32-S3 interfacing with two independent axes over a shared I2C bus. Each axis is equipped with a Time-of-Flight sensor, a servo motor, and a limit switch. During docking, the ToF sensor detects when the arm clears the trolley boundary by registering a sudden range increase, triggering the servo to flip a mechanical block into its engaged position. The belt then reverses, retracting both arms until the limit switches confirm contact with the trolley frame, establishing a secure coupling.

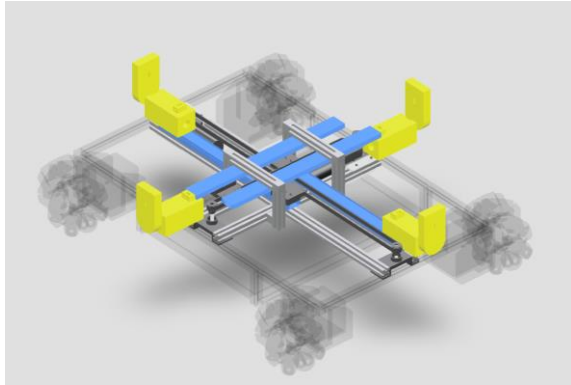


Figure 3: Belt Extension and Clamping System CAD Diagram

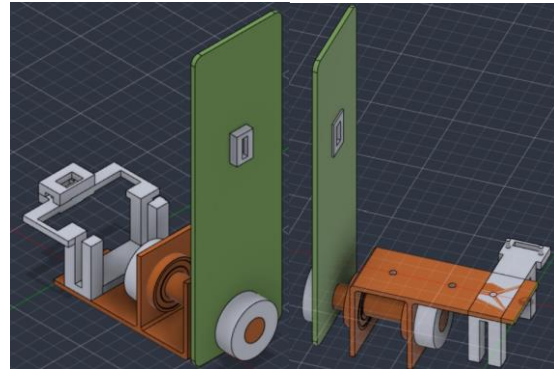


Figure 4 and 5: Flipping mechanisms

3) Electronics & Motion Control System

The electronics architecture is a two-controller distributed system built on the staged power tree of Fig. 6, which separates the 36 V main power system from a USB-fed logic system joined to it at a single point. Three custom PCBs were designed and fabricated for it: a base Wheel Drive board (Appendix A), an Arm Subsystem board (Appendix B), and four identical Driver adapter boards (Appendix C).

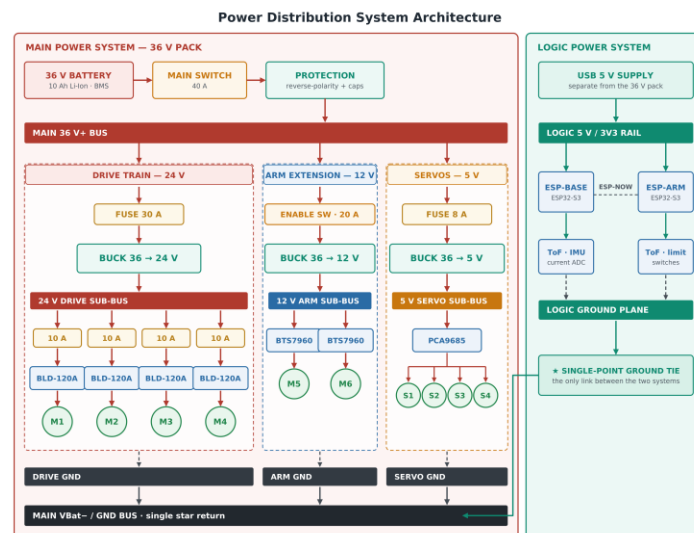


Figure 6: Power distribution system architecture diagram

A custom 36 V, 10 Ah lithium-ion pack with integral BMS feeds a 40 A main switch and a reverse-polarity and bulk-capacitance stage supplying the main 36 V bus. Three independently protected branches descend from that bus: a 30 A fuse ahead of the 36→24 V converter feeding the four BLD-120A brushless drivers; a 20 A latching enable switch ahead of the 36→12 V converter feeding the two BTS7960 H-bridges and arm-extension motors M5–M6; and an 8 A fuse ahead of the 36→5 V converter feeding the PCA9685 expander and four GX3345BLS

flipper servos S1–S4 on the arm board. Each driver carries a further 10 A fuse on the 24 V sub-bus, so a single motor fault clears locally, and each driver's P-sv overload trim is set to the motor's rated 5 A (120 W at 24 V). Gating the 12 V branch at its converter input also staggers inrush: all three converters charging together latched the pack BMS, whereas energising the third separately keeps the transient below it. The drivers' optocoupler inputs idle near 5 V, so each control line is switched by a 2N3904 on its Driver board, whose 100 kΩ pull-down holds every motor disabled while a controller is unpowered or in reset. Each branch returns through its own ground to the main VBat– star bus, meeting the logic ground plane at the single tie of Fig. 6; without that bond, made at the Wheel Drive board's J3 terminal, the base controller does not boot.

III. MODELLING

Analytical models were developed to validate component selection and predict system performance against the functional requirements defined in Table 1. Three critical domains were modelled: drivetrain sizing, structural integrity, and thermal management.

A. Component Selection

Motor and Gearbox Selection: The drivetrain must move a gross system mass of up to 125 kg (100 kg trolley + load, plus ~25 kg AMR) at a target speed of 0.5 m/s on flat indoor surfaces. For a 152 mm diameter wheel (radius $r = 0.076$ m), the required wheel speed is:

$$\omega = v / r = 0.5 / 0.076 \approx 6.6 \text{ rad/s} \approx 63 \text{ RPM}$$

Assuming a worst-case rolling resistance coefficient of $\mu = 0.015$ for rubber on smooth concrete and a 15° castor misalignment drag factor, the total tractive force required across four wheels is approximately 40–60 N. Each motor must therefore deliver at least 15 N of continuous force at the wheel. With a 15:1 planetary gearbox and a wheel radius of 0.076 m, the motor torque requirement at the gearbox input is:

$$\tau_{motor} = (F \times r) / (\eta \times G) = (15 \times 0.076) / (0.88 \times 15) \approx 0.086 \text{ Nm}$$

The BGM 80 mm 120W BLDC motor delivers approximately 0.38 Nm at rated speed, providing a 4.4× torque margin that accounts for acceleration transients, ramp gradients, and the worst-case castor misalignment condition specified in Table 1. The variables used are summarised in Table 2.

Variable	Symbol	Value	Unit	Source
Gross system mass	m	125	kg	Trolley (100) + AMR (25)
Target travel speed	v	0.5	m/s	System requirement
Wheel radius	r	0.076	m	152 mm mecanum wheel
Gearbox ratio	G	15	—	BGM motor spec
Gearbox efficiency	η	0.88	—	2-stage planetary estimate
Rolling resistance coeff.	μ	0.015	—	Rubber on concrete
Motor rated power	P	120	W	BGM datasheet
Motor rated torque		0.38	Nm	BGM datasheet

TABLE 2: Drivetrain Model Variables

Battery Sizing: The average system power draw was estimated at 240W (four motors at ~50% average load plus 40W electronics overhead). For one-hour minimum runtime with a 20% depth-of-discharge reserve:

$$C = P \times t / (V \times \text{DoD}) = 240 \times 1 / (36 \times 0.8) = 8.3 \text{ Ah}$$

Peak current draw at 36V was calculated from an unlikely worst-case scenario of all components drawing maximum power and determined to be 20A.

$$\text{Max power draw} = 4 \times 120\text{W (drive motors)} + 2 \times 120\text{W (arm motors)} + 4 \times 25\text{W (Servos)} = \underline{820\text{W}}$$

$$\text{Max current draw at 36V} = 820 / 36 = \underline{22.8\text{A}}$$

A 36V 10Ah Li-Ion pack capable of delivering 2 – 3 C (20 – 30A) discharge rate was selected.

Thermal Analysis: At 75% motor efficiency, each 120W motor dissipates approximately 40W of waste heat. An 80 mm flange motor in still air can naturally convect 15–25W. By mounting the BLD-120A controllers directly to the aluminium extrusion frame, the chassis acts as a passive heatsink, raising the effective dissipation capacity to ~50W per motor. Motor case temperature is predicted to remain below 80°C at 25°C ambient during continuous one-hour operation, eliminating the need for active cooling.

B. Performance Predictions

Motor Bracket Structural Performance: The D1/D2 motor bracket was analysed for three failure modes: bending of the flat bar plate, cantilever deflection of the D1 rails, and T-nut joint slip. All components exceed a 16× safety factor under the 3 kg motor and gearbox load, as shown in Table 3.

Component	Failure Load (kg)	Motor Load (kg)	Safety Factor
Flat bar plate (20 × 5 mm)	82	3	27×
D1 rail (cantilever, each)	86	1.5	57×
D1 pair combined	172	3	57×
T-nut joints (4 per motor)	50	3	16×

TABLE 3: Motor Bracket Structural Safety Factors

Height Compliance: The total AMR height from ground to the top of the main frame is approximately 130 mm, well within the ≤200 mm trolley clearance requirement specified in Table 1. The 80 mm suspended battery shelf sits between the wheel axle line and the main frame, with 22 mm ground clearance maintained.

Clamping Force: The synchronous belt extension system must withstand a minimum holding force of 300 N as specified in Table 1. The HTD3M belt and servo-actuated flip block design transfers the clamping load through belt tension and guide rail reaction forces rather than through the servo itself, which only needs to overcome the block's rotational inertia during the flip action.

Clamping Force Analysis: The clamping system must withstand a minimum holding force of 300 N (≈30 kg) as specified in Table 1. The force path through the mechanism involves three distinct phases, each with different loading conditions.

Phase 1 — Arm Extension (belt in tension): During extension, the belt only needs to overcome the arm's sliding friction along the guide rail. Assuming a linear rail friction coefficient of $\mu_{\text{rail}} = 0.2$ and an arm mass of $m_{\text{arm}} \approx 0.5$ kg:

$$F_{\text{extend}} = \mu_{\text{rail}} \times m_{\text{arm}} \times g = 0.2 \times 0.5 \times 9.81 \approx 1.0 \text{ N}$$

A standard 10 mm-wide HTD3M belt has a rated working tension of approximately 60 N, providing a 60× margin for the extension phase.

Phase 2 — Block Flip (servo torque): When the ToF sensor triggers, the servo rotates the flip block 90° into its engaged position. The torque required to flip the block is governed by its mass and moment arm:

$$\tau_{\text{flip}} = m_{\text{block}} \times g \times d = 0.05 \times 9.81 \times 0.01 \approx 0.005 \text{ Nm}$$

A standard SG90 micro servo delivers 0.18 Nm at 4.8V, providing a 36× torque margin. The flip action is a one-shot rotation against gravity only — no external load acts on the block during the flip.

Phase 3 — Holding (static load, belt disengaged): Once the arms retract and the limit switches confirm contact, the flip block hooks over the trolley frame edge. The 300 N holding force is transferred through the following structural path:

Trolley frame → Flip block (shear) → Arm body → Guide rail → Chassis frame

Critically, this load path bypasses the belt entirely — the guide rail and its mounting hardware carry the shear force, not the belt tension. The flip block, assuming a minimum cross-section of 15 × 15 mm in aluminium 6061-T6 (shear strength ≈ 207 MPa), can withstand:

$$F_{\text{shear}} = \tau \times A = 207 \times (15 \times 15) = 46,575 \text{ N}$$

This provides a safety factor exceeding 150× against the 300 N requirement. The guide rail mounting bolts (M4, grade 8.8, two per rail) each handle approximately 3,400 N in shear, providing a combined capacity of 6,800 N per arm at 22× safety factor.

The analysis confirms the system meets the 300 N holding force requirement with substantial margins at every point in the load chain. The weakest link is the guide rail mounting bolts at 22×, which remains well above acceptable engineering practice. The variables used in this analysis are summarised in Table 4.

Variable	Symbol	Value	Unit	Safety Factor
Required holding force	F_hold	300	N	—
Arm mass	m_arm	0.5	kg	—
Guide rail friction coeff.	μ_{rail}	0.2	—	—
GT2 belt working tension	F_belt	60	N	60× (Phase 1)
Flip block mass	m_block	0.05	kg	—
Flip block moment arm	d	0.01	m	—
Servo rated torque (SG90)	τ_{servo}	0.18	Nm	36× (Phase 2)
Block cross-section	A	225	mm ²	150× (Phase 3)
Alu 6061-T6 shear strength	τ_{shear}	207	MPa	—
Rail bolt capacity (2× M4)	F_bolt	6,800	N	22× (Phase 3)

TABLE 4: Clamping Force Model Variables

IV. RESULTS

A. Prototypes

Downscaled Control Prototype: Using L298N drivers, ToF sensors and 6V DC motors with mecanum wheels were used.

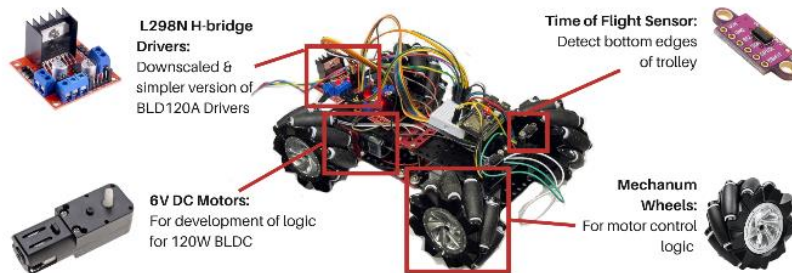


Figure 7: Mecanum wheel drive prototype

Full-Scale Prototype: Built with the 2020 extrusion chassis, BGM motors with BLD-120A drivers, mecanum wheels, belt extension system.



Figure 8 and 9: CAD diagram of Full Prototype and actual Prototype

B. Experiments

Experiments	Objectives	Findings
Castor wheel position detection and manipulation	Time to adjust and move in desired direction to be < 3s	It was able to do so
Holding Force of Arm Extension	Final yield weight measured on scale to be > 30kg	It was able to do so
Time to dock and clamp	Time to alignment < 60s	Done within 60s
Movement speed	Time taken to cover distance to be at average speed 0.5-1.0 m/s	It was able to do so
Clamp reliability	Have 50 successful cycles	It was able to do so

TABLE 6: Experiments Objectives and Findings

V. DISCUSSION

Beyond simply moving items from point A to point B, introducing this AMR fundamentally changes how a workspace functions day-to-day. In busy textile facilities or cramped manufacturing floors, space is usually at a premium, and traditional automated guided vehicles often cause bottlenecks due to wide turning radii. By using omnidirectional movement, this system flows naturally with human traffic and tight layouts rather than fighting against them. The biggest real-world shift is how it handles transport integration. The AMR drives right underneath existing factory trolleys, couples automatically, and moves out. This creates a seamless loop between storage and assembly lines,

taking the physical strain off workers, cutting down on transit delays, and drastically reducing the risk of workplace injuries or dropped materials.

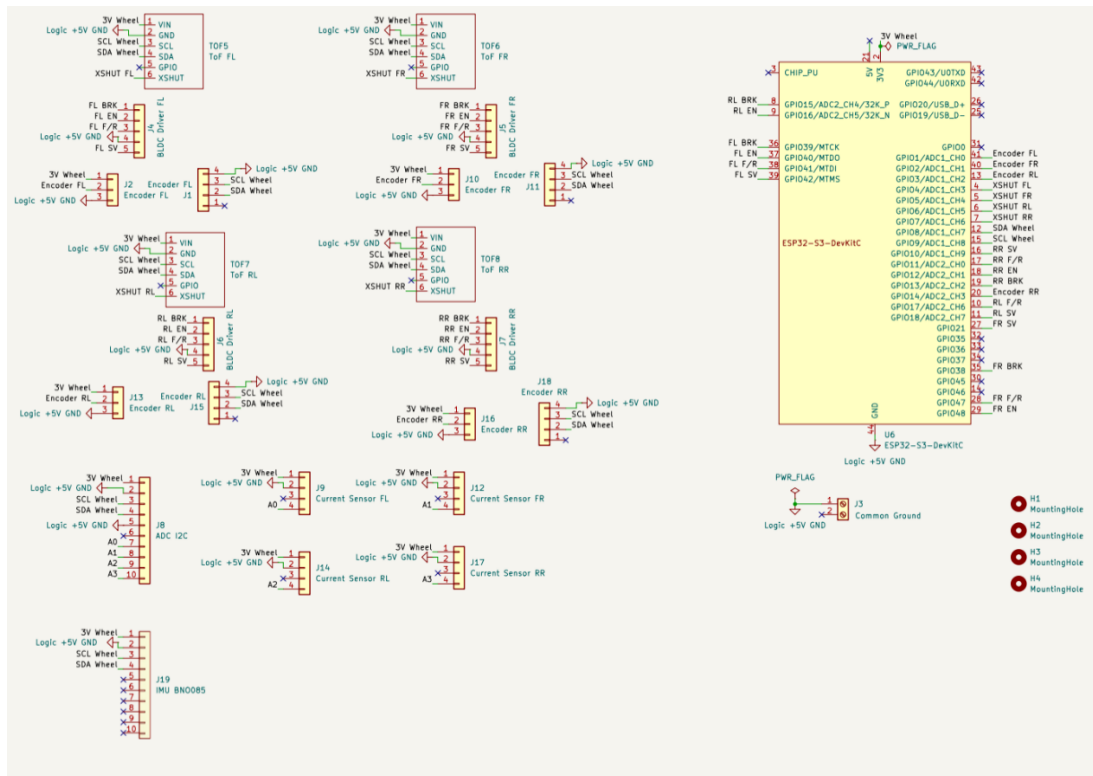
This project proves that reliable automation doesn't require a massive corporate budget or proprietary, five-figure systems. By combining accessible components with rigorous engineering, we've built a high-performance AMR from the ground up, shifting the paradigm from rigid, off-the-shelf products to robust, locally adaptable robotics.

VI. CONCLUSIONS

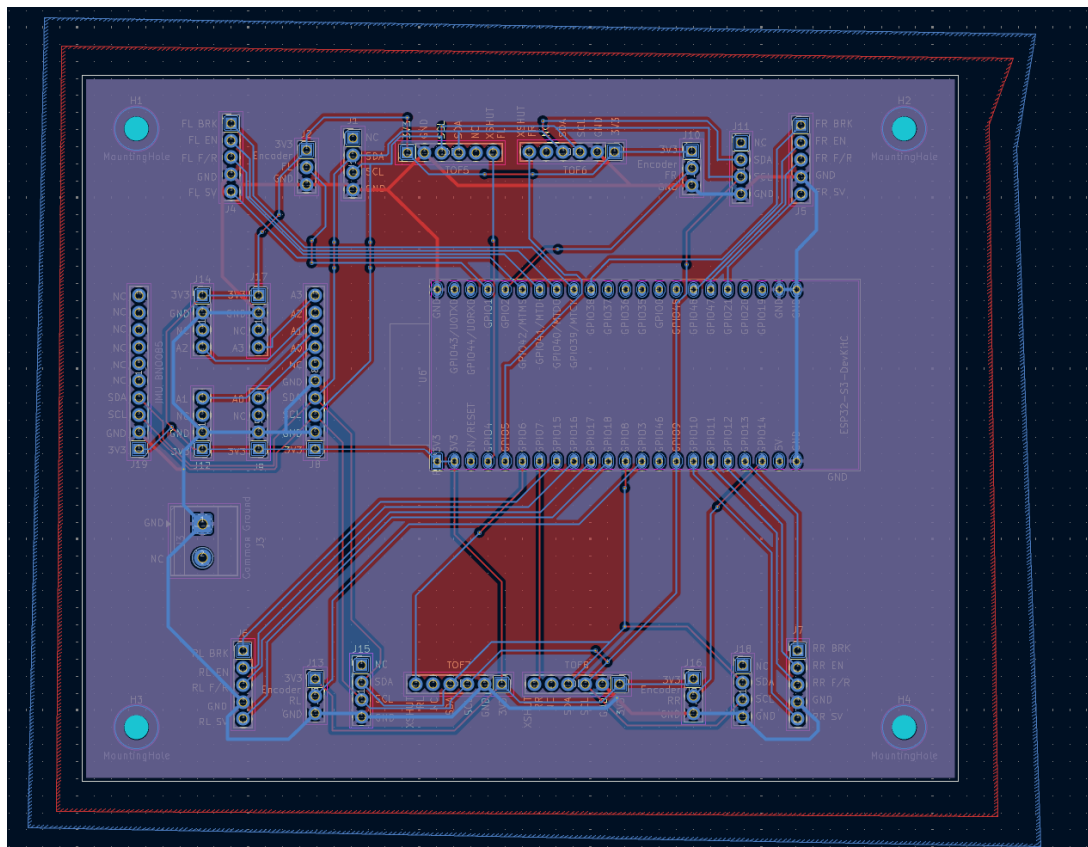
This project establishes a viable, engineering-driven framework for democratizing internal logistics automation. By addressing confined workspaces with a bespoke mecanum drive architecture and integrated coupling mechanisms, we move beyond the limits of rigid commercial platforms. Rigorous mechanical modeling and structural validation confirm that high-performance, load-bearing robotics can be achieved using accessible, modular construction methodologies.

Looking ahead, this foundational platform paves the way for advanced development and scalability. Future work will focus on integrating SLAM algorithms for dynamic navigation, implementing centralized fleet management for multi-robot synchronization, and expanding end-effector adaptability. Ultimately, this offers a scalable blueprint for localized automation, proving that agile, cost-effective robotics can be systematically engineered for real-world operational demands.

APPENDIX A

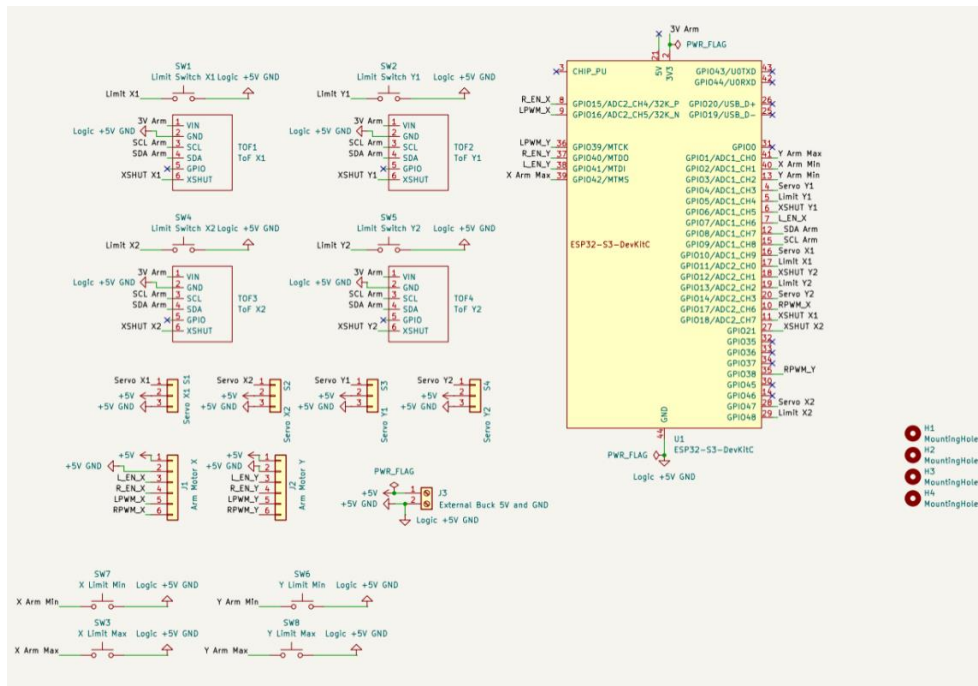


Wheel Drive Schematic

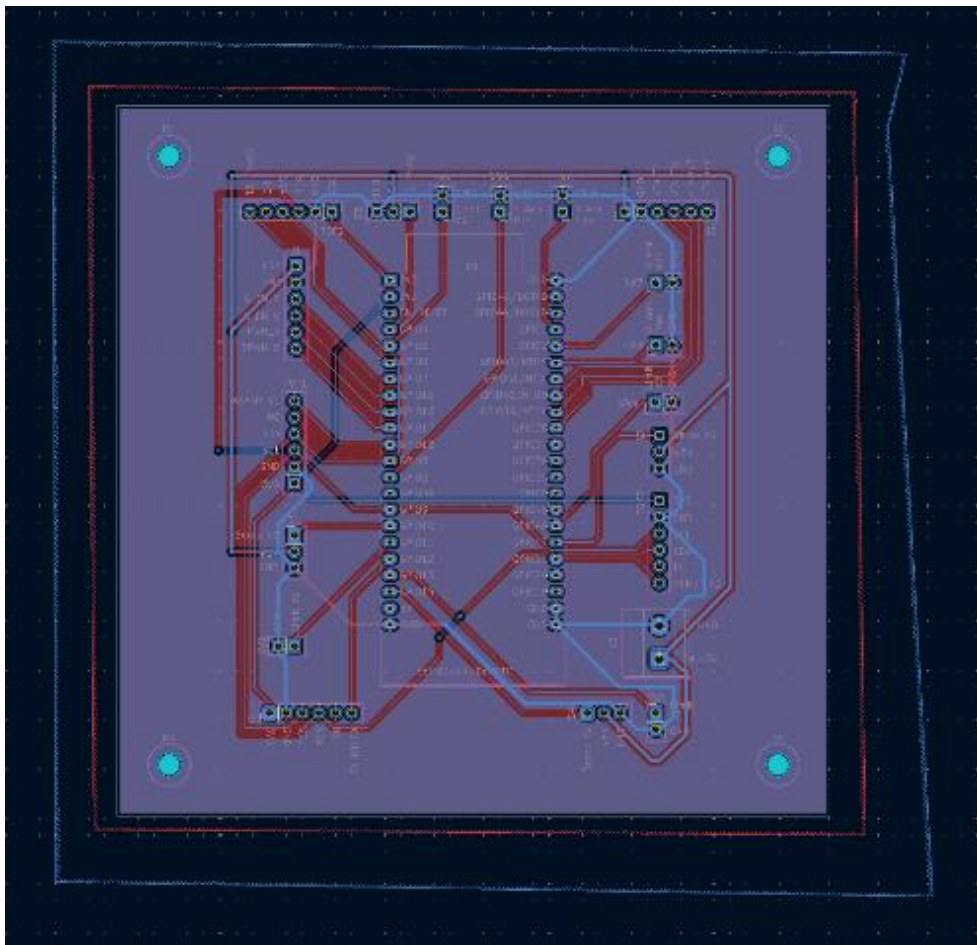


Wheel Drive PCB

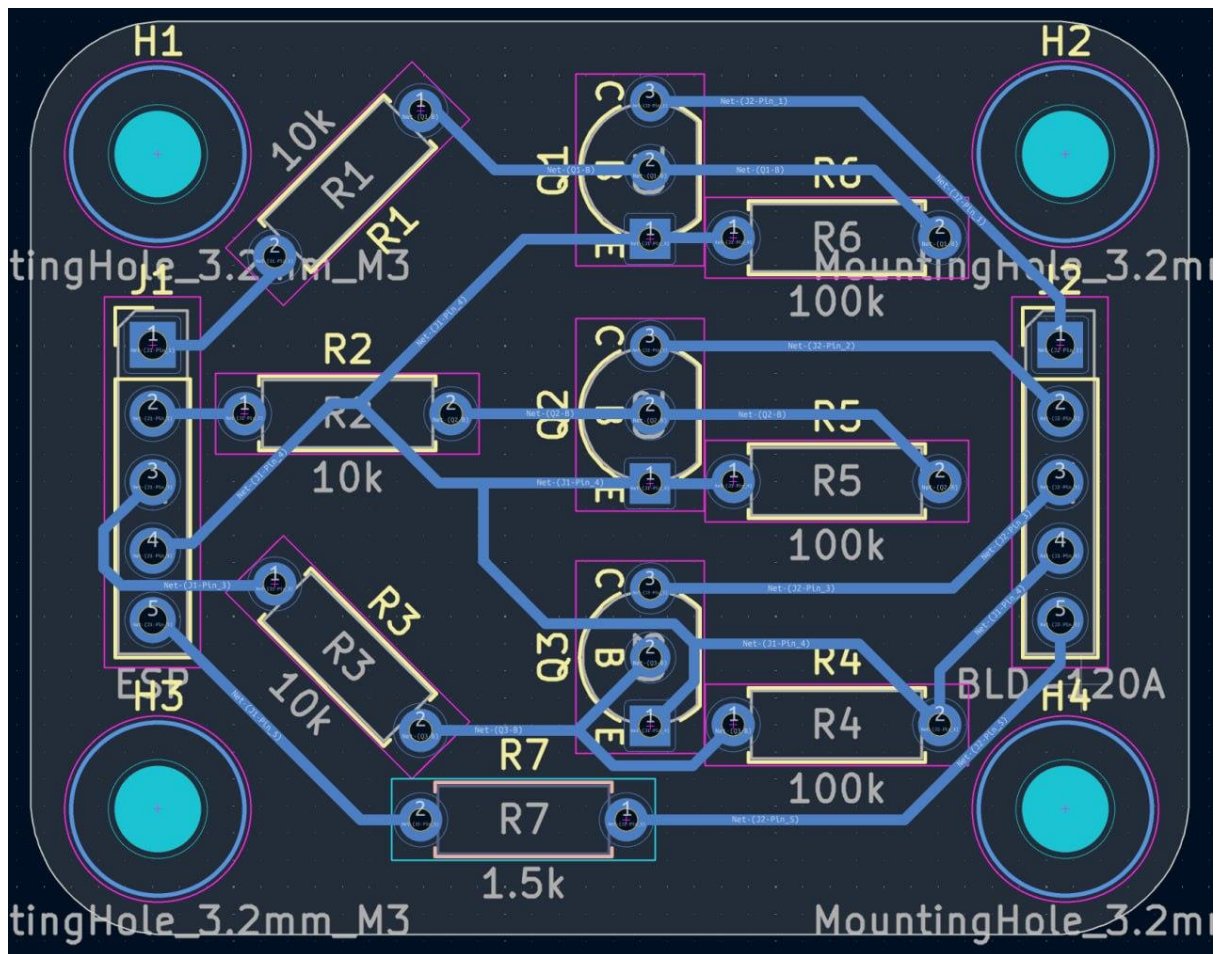
APPENDIX B



Arm Subsystem Schematic



Arm Subsystem PCB



Driver adapter board

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